

Biocompatibility, biodegradability, toxicity, life cycle analysis, social economic aspects, environmental, and health impact of biopolymers

**Meriem Kasbaji¹, Fatima Ezzahra Atmani⁴, Mohamed Amine Kasbaji²,
Nabil Grimi⁵, Mohamed M'Barki², Mounir El Achaby¹, Amine Moubarik³ and
Mustapha Oubenali²**

¹*Materials Science, Energy and Nanoengineering Department (MSN), Mohammed VI Polytechnic University, Ben Guerir, Morocco,*

²*Team of Analytical & Computational Chemistry, Nanotechnology and Environment, Faculty of Science and Technologies, Sultan Moulay Slimane University, Beni Mellal, Morocco,*

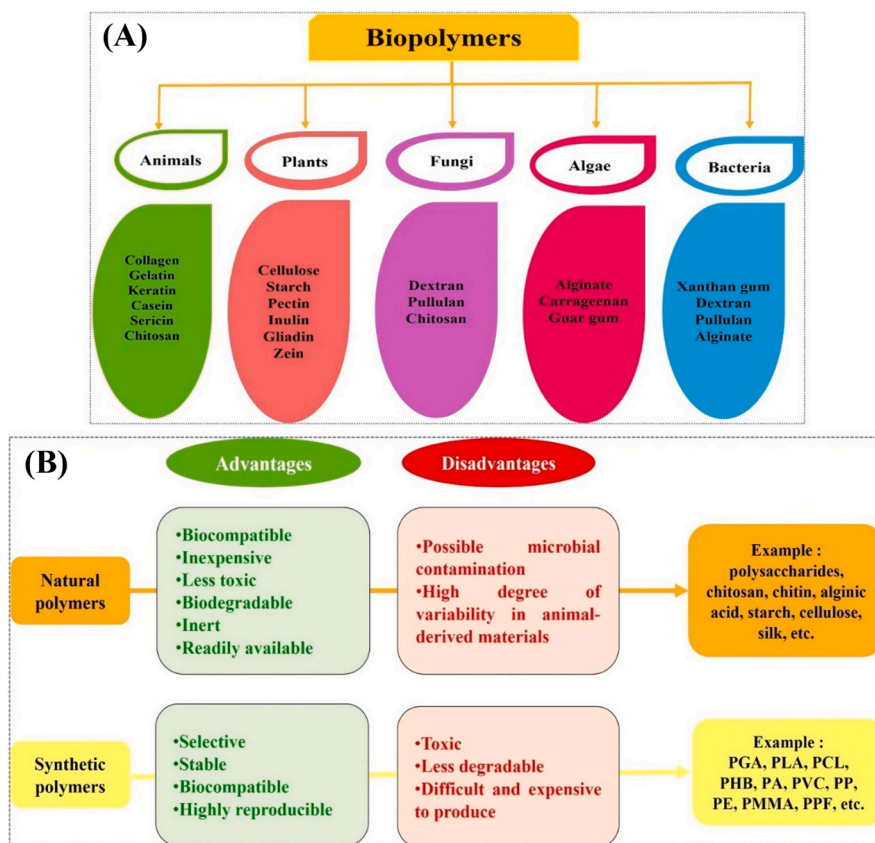
³*Chemical Processes and Applied Materials Laboratory, Polydisciplinary Faculty, Sultan Moulay Slimane University, Beni Mellal, Morocco,*

⁴*Laboratory of Environmental, Ecological and Agro-industrial Engineering, Faculty of Science and Technologies, Sultan Moulay Slimane University, Beni Mellal, Morocco,*

⁵*Sorbonne Université, Université de Technologie de Compiègne, Laboratoire Transformations Intégrées de la Matière Renouvelable (UTC/ESCOM, EA 4297 TIMR), Centre de recherches Royallieu, Compiègne Cedex, France*

3.1 Introduction

Throughout history, humans have primarily used biopolymers to produce food and clothing. However, since the Industrial Revolution, fossil fuels, particularly oil, have become a critical energy source to produce commercial products, such as plastic. Nonetheless, the limited supply of fossil fuels and the environmental impact of their use necessitates a long-term strategy that considers the planet's natural cycles. As a result, there is renewed interest in biopolymers, which are renewable resources (Baranwal et al., 2022; Deniz et al., 2008). Biopolymers are polymers produced from natural sources like animals, microorganisms, agricultural wastes, and plants, which comprise various polysaccharides, and polypeptides (Fig. 3.1A)

**FIGURE 3.1**

(A) Classification of biopolymers based on their origin and (B) advantages and disadvantages of natural and synthetic polymers.

(Muir & Burdick, 2021; Park et al., 2017). These can be synthesized chemically from biological components or biosynthesized by living organisms (Smith et al., 2016). Biopolymers are classified into three types based on their origin: (1) polymers derived from microbial or animal products that may be useful in medical and pharmaceutical practices, (2) biodegradable polymers synthesized chemically from petrochemical resources or modified from natural polymers, and (3) polymers produced from renewable resources/biomass, such as agro-polymers derived from agricultural resources (Primožič, 2021).

Biopolymers are biological macromolecules that are created through the polymerization of monomers, that are covalently linked together (Pathak et al.,

2022; Patra, 2018). These monomeric particles combine to produce larger molecules (Ashter, 2016). Biopolymers have been discovered to be biocompatible, biodegradable, renewable materials, nontoxicity, and ability to adsorb bioactive molecules, making them valuable for a variety of applications in the food and pharmaceutical industries, including edible films, emulsions, and packaging materials in the food industry, as well as drug transport materials, medical implants, wound healing, tissue scaffolds, and dressing materials in the pharmaceutical industry (Baranwal et al., 2022). Biodegradable biopolymers are becoming increasingly important, with ongoing research focusing on developing newer variations. A biopolymer made from every year renewable resources that biodegrades appears to have the potential to solve the major problems associated with plastics. The raw material is no longer of fossil origin, improving resource security, and the polymer can biodegrade to leave no waste product at the final stage of its existence (Brown, 2004). If a biopolymer ends up in a landfill and does not get the light, water (H_2O), oxygen, and temperature it needs to break down, it might not break down at all (Polman et al., 2021). A diverse range of these polymers have been synthesized or occur naturally during the growth phases of various organisms. Fig. 3.1B schematically depicts the advantages and disadvantages of synthetic and natural polymers (Baranwal et al., 2022; Das et al., 2023; Maghsoudi et al., 2020). Certain microorganisms and enzymes have been discovered that can break them down. Various classifications for biodegradable polymers have been proposed as synthesis techniques evolve (U.S. Congress, 1993).

Biocompatible polymers are used in the synthesis of nanocomposites that have changed the world of medicine by innovating next-generation materials. These nanocomposites have enormous potential for developing highly efficient smart materials (Pathak et al., 2023). The widespread use of synthetic and chemical-based polymers in the food and medical industries raises several environmental concerns. Raised awareness of environmental issues, pollution controls, and municipal solid waste disposal are all driving forces in the development of biopolymer-based packaging materials (Siracusa et al., 2008). These resources are renewable and possess remarkable attributes, such as low weight and density, low prices, and quick production. Furthermore, because natural polymers do not pollute land or H_2O and are processed more slowly than synthetic ones, they also cause less damage to machinery and its components (Vinod et al., 2020). By using biopolymers, carbon dioxide emissions, waste from municipal solid sources, and dependency on petroleum-based resources are reduced (Peelman et al., 2013). Researchers have become increasingly interested in bio-based materials, such as fibers, biopolymers, and bio-composites because of increased environmental awareness over the past few decades. These materials will eventually replace synthetic materials (Liu et al., 2019; Mohan & Kanny, 2019). In the future, biopolymers will place a huge demand on producers of innovative materials. However, the materials' cost-effectiveness needs to increase because they are being donated specifically for sustainable development. These polymers properties ought to be put to use in new material applications, and goods ought to be made with those characteristics in mind. As a

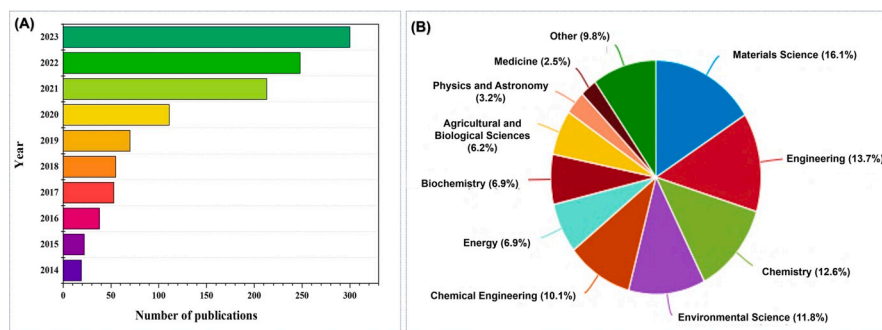
result, they are starting to emerge and are taking greater responsibility for protecting the planet we live (Rao et al., 2014; Yadav et al., 2015). Biopolymers, due to their diversity and versatility, have the potential to be used in a wide range of industries. Adhesives, absorbents, lubricants, soil conditioners, cosmetics, drug delivery vehicles, textiles, high-strength structural materials, and even computational switching devices can be made from them. While many biopolymers are still in the development stage, significant applications in packaging, food production, and medicine are beginning to emerge. Some biopolymers have the potential to directly replace synthetic materials in traditional applications, while others have unique properties that could open up new commercial opportunities (Bang & Suslick, 2010; Li et al., 2020). However, using these materials raises several issues and difficulties because their sustainability is not implied by the fact that they are made from renewable raw materials. Consequently, research must be done to determine the possible environmental effects of the newly developed biopolymers, for which the life cycle assessment (LCA) method can be used (Giovanna Mercado et al., 2017). First, this chapter provides a quick overview of the key ideas behind biopolymers and then the distinction between biopolymer's biodegradability and biocompatibility. Following that, the life cycle analysis, toxicity, and socio-economic aspects of biopolymers will be discussed. The final section discusses the effects of biopolymers on the environment and human health (Fig. 3.1).

3.2 Biopolymers: an overview

3.2.1 Definition and applications

The term “biopolymer” originally derives from the Greek words “bio” and “polymer,” referring to the nature, concept, and different types of living organisms. Because of their durability, biodegradability, and biocompatibility, biopolymers are a sustainable and viable alternative to most conventional synthetic polymers. Renewable, sustainable biological sources, including plants, animals, and micro-organisms, provide the origin of several biopolymers.

In recent years, science has taken a keen interest in all things “green.” Many researchers have focused on the development of materials based on natural, biodegradable processes and sources, with the aim of replacing nonrenewable energy sources in the context of sustainable development. The number of scientific publications relating to this change of perspective is illustrated in Fig. 3.2A and B. The number of publications has increased in recent years in several areas. Due to their interesting properties, biopolymers are widely used in many sectors, including food packaging, bioengineering materials, wastewater treatment, anticorrosive agents, and many other areas (Baranwal et al., 2022; Gopinath et al., 2023). Over the last 10 years, there has been growing interest in various industrial and academic fields, including nanotechnology, for which nanocomposites derived from

**FIGURE 3.2**

(A) Number of scientific documents on “biopolymer” in the last 10 years as per the ScienceDirect survey (*date of search: December 28, 2023*) and (B) contributions of biopolymers scientific papers by subject area.

biopolymers and polymers of nanometric dimensions and size have been developed. In addition, great importance and attention have been given to the development of natural biopolymers, whose qualities are essentially based on their size, concentration, and structure. To exploit the full potential of high-performance, adaptable biopolymers, researchers are continuing to carry out studies and research in this context (Gheorghita et al., 2021; Jain et al., 2021; Kasbaji Mennani, Oubenali, et al., 2023) (Fig. 3.2).

Natural biopolymers, such as cellulose, chitosan, lignin, silk fiber, and many others contribute significantly to all areas of the life cycle. In addition to other biotechnological materials, biopolymers are also used in the manufacture and construction of buildings and structures due to their inherent biodegradability as well as their superb biocompatibility. They were initially considered as packaging materials in the pharmaceutical industry, for encapsulating drugs, dispensing medicines, and probiotic agents (Jain et al., 2021; Phiri et al., 2023; Plank, 2004). Moreover, as a result of the evolution and development of several types of technology, these materials are applied in a wide range of sectors, such as agriculture (Phiri et al., 2023), the automotive industry (Yusuf et al., 2024), food packaging (Rahman et al., 2023), drug distribution (Sharma et al., 2024), biomedicine (Lu et al., 2023), biosensing devices (Rodoshi et al., 2024), wastewater purification (Kasbaji, Mennani, Boussetta, et al., 2022; Kasbaji, Mennani, Grimi, et al., 2023), corrosion inhibitors (Chauhan et al., 2023), packaging and tissue technology (Grzebieniarz et al., 2023), environmental protection (Chaudhari et al., 2023), and so on. Some biopolymers naturally possess antibacterial and antioxidant capabilities while others do not exhibit these properties in their natural state (Lu et al., 2023; Sharma et al., 2024). In addition, biopolymers are increasingly used in electrical activities, where they can potentially reinforce electronic systems. Moreover, these materials are the source of recent technologies known as organic

transient bioelectronics. These materials are also being applied as electrolytes and separators to solve certain battery-related challenges and problems (Xu et al., 2022). Recently, this type of biopolymer has been introduced into the manufacture of nanomaterials, as well as into 3D printing processes, which represent a new technique due to their promising application (Kasbaji, Ibrahim, et al., 2023; Kasbaji, Mennani, Grimi, et al., 2022; Li et al., 2021).

3.2.2 Biopolymer: types and sources

Biopolymers refer to all natural polymers present in living organisms, such as plants, animals, microbes, and other natural sources and substances. According to their origin and nature, biopolymers are essentially classified into two categories: natural biopolymers and synthetic biopolymers, as illustrated in Fig. 3.3A. The different types of biopolymers as well as their application and sources are summarized in Table 3.1.

3.2.2.1 Natural biopolymer

The term “natural biopolymer” refers to the formation of an organic biopolymer from living organisms. The category of natural biopolymers includes proteins and polysaccharides (Subash et al., 2023).

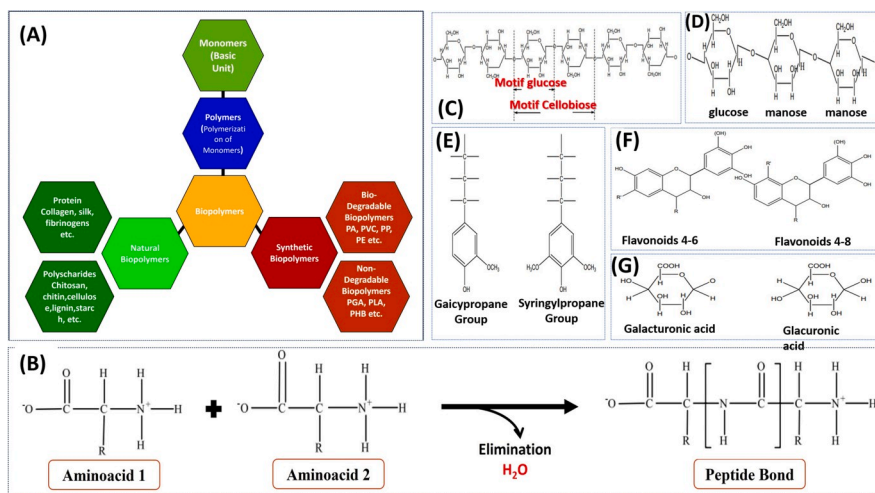


FIGURE 3.3

(A) Classification and types of biopolymers; (B) formation of peptide bonds in biomolecular proteins; (C) chains cellulosic; (D) examples of hemicellulose chains: glucomannose; (E) example of lignin monomer groups; (F) chemical structure of tannins; and (G) example of the chemical structure of two uronic acids.

Table 3.1 Various types of biopolymers, their technical characteristics, and applications.

Biopolymers	Properties	Application	References
Polysaccharides	Reduced water content in cement, excellent mechanical and bonding performance	Cosmetics; used as additives; structural materials in plants and some higher organisms (cellulose, chitin); energy storage materials (starch, glycogen); and molecular recognition (blood types)	Chen et al. (2023)
Lignin	Increase workability, Insulating material and protective coatings	Cosmetics; drug delivery systems; packaging; sensors; wastewater treatment; wearable electronics	Mennani et al. (2024)
Proteins	Antioxidant; antihypertensive	Biological catalysts (enzymes), growth factors, receptors, structural materials (wool, leather, silk, hair, and connective tissue), hormones (insulin); toxins, and antibodies	Reddy and Yang (2011) , Ihsan et al. (2024)
Microbes	Self-healing concrete repairs, reduces water permeability	Effective remediation technique of crack repair	Noor Azammi et al. (2019)
Cross-linked	Improved mechanical strength	Used as admixtures in concrete; packaging, and medical sector	Tan et al. (2023)
Xanthan gum	Foam stabilizer	Beer, retarder, pore filler, resins, packaging, and medical sector	Bhat et al. (2022)
Arabic gum	Improved mechanical properties of concrete, building process and resins sector	Used as superplasticizer, retarder, pore filler, crack bridging agent, and strength enhancer	Prasad et al. (2022)

3.2.2.1.1 Protein

Proteins are very large and complex molecules that perform several major functions in the human body. They are made up of several hundred or thousands of smaller components, known as amino acids, linked together by very long chains. They perform most of the tasks in cells and are essential for the formation, function, and maintenance of the body's structures, tissues, and organs. Each protein is made up of 20 distinct types of amino acids, which can combine with each other in a variety

of ways. Vegetable-based proteins, such as those contained in food products including wheat, corn, and soy, have a higher immunogenicity than those of animal origin. They also have a low molecular weight and a high concentration of net negative charges, making them effective drug carriers. Indeed, the hydrophilic nature of proteins is explained by the high presence of polar amino acids, which are perfectly suited to attracting cells. Natural biopolymers, such as collagen, silk, and fibrinogen can also be used; collagen sold commercially is used in injections for dermatological, antibody, and immunological purposes. Proteins are mainly structured by peptide bonds and disulfide bonds. A peptide bond occurs when the carboxyl group of one molecule combines with the amine group of the second molecule, releasing a molecule of H_2O . This condensation process (also referred to as dehydration reaction) occurs between multiple amino acids. Fig. 3.3B illustrates the process and mechanism of peptide bond formation (Ihsan et al., 2024; Reddy & Yang, 2011).

3.2.2.1.2 Polysaccharide

In general, polysaccharides are extended chains of monosaccharides connected by glycosidic bonds, normally used by our bodies as a source of energy or as a support for cell structure. Examples of natural polysaccharides include chitosan, chitin, cellulose, lignin, hemicellulose, alginic acid, starch, cellulose, konjac, and gums (guar gum, gum karaya, etc.). In addition, chitosan is a natural polysaccharide that can be used extensively in drug management and tissue technology. In the agri-food sector, alginic acids can be used extensively in the production and processing of food products and commodities. Starch, for its part, is a classic and typical natural polysaccharide, used as a coating and adhesive due to its insolubility in H_2O (Mohanta et al., 2023; Subash et al., 2023). Cellulose is a multifunctional element in many types of food, fulfilling functional tasks, such as thickening, gelling, and texture enhancement. In low-calorie foods, cellulose can act as a fat substitute, emulsifier, and bulking agent (Subash et al., 2023). These structural and textural properties make cellulose an element of choice in food processing and other important sectors as the effluent treatment sector (Kasbaji, Mennani, Grimi, et al., 2023; Kausar et al., 2022). Fig. 3.3C–G presents the different structures of several types of polysaccharides materials.

3.2.2.2 Synthetic biopolymers

Synthetic biopolymers are a type of polymer modified based on natural polymers or chemically manufactured using synthetic monomers so that they can degrade naturally without producing waste harmful to living organisms and the ecosystem. For some years now, synthetic biopolymers have enjoyed numerous advantages over natural polymers in terms of stability and flexibility, enabling them to cover a wide range of applications. Nevertheless, natural biopolymers are favored over synthetic polymers thanks to their biodegradability and environmental safety (Shamsuddin et al., 2018). Due to advances in molecular development tools and

polymer chemistry, biopolymer synthesis can now be tailored to specific applications. Because of some of their outstanding characteristics, such as stability, regulated release, nonimmunogenicity, and elimination from the body, synthetic biopolymers are among the most common applications in the medical industry, corresponding to their application in the human body. Compared with ceramic and/or metal particles, artificial biopolymers are easier to produce in a variety of shapes and sizes. These can be used in industry, with low production costs, making them vitally important in our daily lives. The process of synthesizing biopolymers into the desired biopolymer molecules is generally carried out by enzymes and living cells. Synthetic biopolymers are generally classified into two broad categories: nonbiodegradable biopolymers and degradable biopolymers (Burzic et al., 2019; Moshood et al., 2022; Shamsuddin et al., 2018).

Nonbiodegradable polymers are characterized by their ability to resist environmental degradation, and therefore end up as waste products. Nonbiodegradable biopolymers have good resistance to H₂O and different climatic conditions and are frequently used for nonbiomedical applications. Common examples include polyamide (PA), polyvinyl chloride (PVC), polypropylene (PP), polyethylene (PE), polymethyl Methacrylate (PMMA), polycarbonate (PC), polyurethane (PU), and others (Dwivedi et al., 2019; Moshood et al., 2022; Peelman et al., 2013).

The term “biodegradable polymers” refers to polymers that degrade when exposed to the elements. The biodegradable nature of biopolymers provides a viable alternative to various petrochemical-based packaging materials. Indeed, these biodegradable synthetic polymers are produced in a controlled environment and possess constant structure and mechanical functions, such as stiffness and modulus of elasticity. Compared with biological scaffolds, their production is less expensive and favors contact with endothelial cells (Vinod et al., 2020). The artificial production of synthetic polymers with biodegradable properties therefore occupies an essential place in cell engineering and other biological fields of application. Synthetic polymers degrade through a simple hydrolysis process (Baranwal et al., 2022). Polyglycolic Acid (PGA), polylactic acid (PLA), polycaprolactone (PCL), polyhydroxybutyrate (PHB), polypropylene Fumarate (PPF), polydioxanone (PDS), etc., are synthetically biodegradable polymers used in pharmaceutical and medical processes. The hydrophilic character of PGA is much higher than that of PLA, leading to higher crystallinity; consequently, this polymer was selected as the fully scalable adsorbable structure initially synthesized (Daphedar et al., 2022).

3.2.2.3 Properties and synthesis methods of biopolymers

Biopolymers represent a type of polymer obtained from biological resources. Given their richness, high strength, high thermal stability, lack of toxicity, antibacterial and antifungal properties, biocompatibility, and biodegradability, as well as exceptional characteristics, such as high adsorption capacity and simplified functionalization, they have been examined in numerous applications in the industrial, biomedical, pharmaceutical, drug delivery, tissue engineering, drug,

tablet, adhesive, paper, food, cotton, and rayon fields, including adsorption process (Fig. 3.4A). Synthetic biopolymers include all polymers modified by a biochemical process. Several processes and techniques are available for biopolymer synthesis. Fig. 3.4B presents some typical techniques for generating effective structural and mechanical characteristics of biopolymers.

3.3 Biocompatibility and biodegradability of biopolymers

3.3.1 Biocompatibility of biopolymers

A biomaterial is typically defined as any substance, excluding food and medications, that can be used over time as part of a system to address or substitute any tissue, organ, or bodily function. Few, if any, materials are completely nonreactive

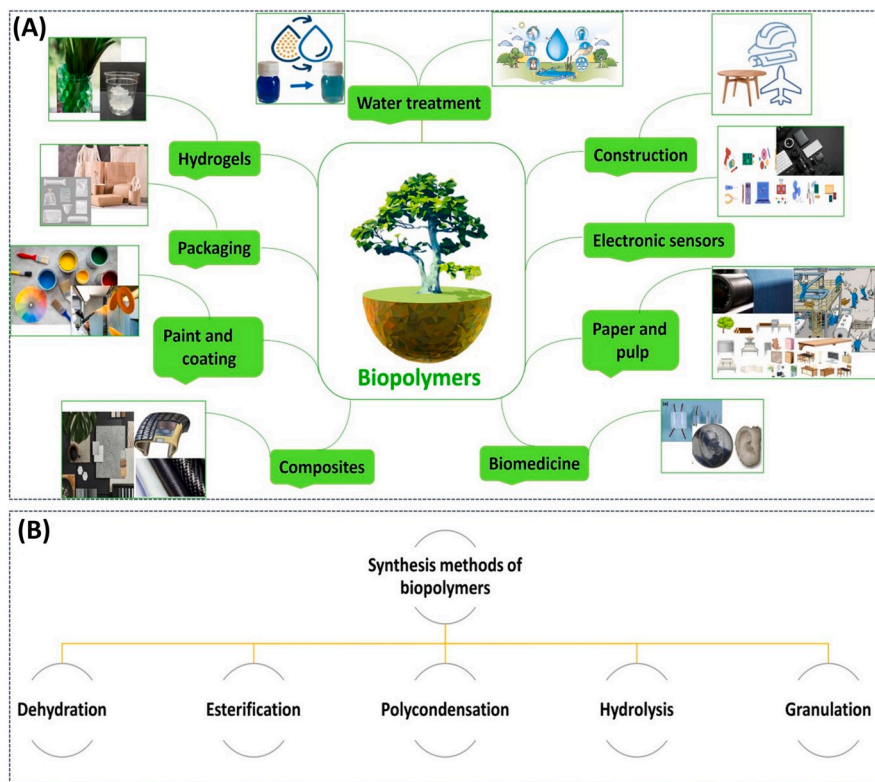


FIGURE 3.4

(A) Various applications for biopolymers in a wide range of industrial sectors and domains and (B) some typical techniques for production of biopolymers.

from a physiological standpoint; the vast majority contain various elements that may be toxic or irritating. Furthermore, chemical reactions during the material's setting process may have harmful effects (Helwig et al., 2003). Before being employed in applications, biomaterials must meet several requirements, one of which is biocompatibility. Its constituent parts ought to go through various tests to validate this feature (Schmalz, 2002).

Ratner defines biocompatibility as the ability of a material to elicit a particular organism's response. Immunological, toxicological, and hemolytic reactions of the materials with living cells are much easier to perform than when they were first tested in animals. However, the main issues are thrombotic and bleeding effects following the insertion of a foreign body into the human body (Ratner et al., 1996). Biocompatibility is further defined as the relationship that exists between a material and an organism in such a way that neither produces undesirable effects. Many works have mentioned biocompatibility, with an increasing interest in evaluating the characteristics of medical and dental materials and devices, as well as the responses caused by their components. Although no ideal pattern for determining these properties has been identified, several methods have been proposed (Jiang et al., 2010). Essentially, the first step in the biocompatibility evaluation process is figuring out what possible biological risks using the material could bring. After risks have been identified, a strategy for assessing them needs to be created (Jorgensen et al., 2021). If a polymer does not cause problems for the body, like allergic reactions or other negative side effects, then it is considered biocompatible (Ambrożewicz et al., 2019). It must be taken into account that the term "biocompatibility" is used broadly to describe the properties of materials that do not cause harm when they come into contact with living organisms, such as proteins, biological cells, and tissues (Tsuruta, 2010). A variety of studies, including in vitro assays and clinical trials, are required to determine a material's biocompatibility. These studies span several disciplines, including pharmaceuticals, biology, chemistry, and toxicology.

3.3.2 Methods for testing biocompatibility

The goal of biocompatibility testing is to develop consistent biological tests for biomaterials to create a more reliable and safe testing process for comparing results across studies. These tests are classified as primary (level I), secondary (level II), and preclinical (level III), and they involve assessing cytotoxicity, irritant potential, and systemic toxicity in animals via intramuscular and subcutaneous implants, as well as observing tissue reactions after material insertion, such as in human teeth. Level I test can be performed in vitro or in vivo. In vitro tests evaluate the material's properties directly in cultured cells that respond to the effects of the experimental material. Many constituents that were initially deemed cytotoxic may be modified or have their use restricted by manufacturers to avoid cytotoxicity. In vivo tests primarily involve implanting materials into subcutaneous or intramuscular areas of

rats and rabbits to assess tissue response to the implanted material after a period of observation (Jiang et al., 2010). Biocompatibility testing includes the following procedures (Bag & Valenzuela, 2017): (1) cytotoxicity (tissue culture), (2) sensitization, (3) irritation, (4) implantation, (5) genotoxicity, (6) hemocompatibility, and (7) preclinical safety testing.

3.3.2.1 Cytotoxicity (tissue culture)

Cell culture assays use isolated cells in vitro to determine the biocompatibility of a substance or extract (Frigaard et al., 2023; Pizzoferrato et al., 1994). These techniques can be used to evaluate the toxicity and irritancy risk of products and chemicals, and the material to be used must be free of cytotoxicity and other harmful effects. This method is excellent for screening materials before subjecting them to in vivo testing (Kirkpatrick et al., 1998, 2005). By definition, the toxicological risks brought on by a substance or its extract in cell culture are referred to as cytotoxicity which is a fast, accurate, and standardized of a material or device (Cao et al., 2005; Zarrintaj et al., 2023). This highly sensitive test is caused using isolated cells in cultures and the absence of physiological effects that are necessary in vivo systems to protect cells inside the body. Cytotoxicity tests can be performed on a variety of cell types (Franz et al., 2009; Zarrintaj et al., 2023).

3.3.2.2 Sensitization

Sensitization tests are used to determine whether a substance contains chemicals that, when used repeatedly or over an extended period, could have negative local or systemic effects. Reactions involving allergies or hypersensitivity are mediated by immune systems. Test material itself, a variety of compounds from the test sample, or, more frequently, extracts from the test material can all be used in studies to evaluate sensitization potential. For all kinds of medical equipment, sensitization testing is advised, per the materials biocompatibility matrix (Humpolicek et al., 2012).

3.3.2.3 Irritation

These studies typically assess the potential for local irritation caused by devices, fabrics, or extracts on the skin or mucous membranes of animals. While it is generally a good idea to slightly overestimate exposure conditions to provide patients with a margin of safety, the length of contact and the route of exposure (skin, eye, or mucosa) should match the anticipated clinical usage of the device (Ramires et al., 2005).

3.3.2.4 Implantation

When medical instruments or biomaterials (such as sutures, surgical ligating clips, implantable devices, etc.) come into close contact with living tissue other than the skin, implant tests are used to evaluate their biocompatibility. Instruments intended for short- or long-term implant in clinical settings will be evaluated in these

evaluations. Utilizing implant techniques, materials that are absorbable and nonabsorbable can be assessed. To provide a reasonable level of protection, the implant trial should closely mimic the intended therapeutic use ([Ramires et al., 2005](#)).

3.3.2.5 *Geno toxicity*

Using a battery of in vitro and in vivo tests, genomic toxicity experiments identify mutagens, molecules that can cause genetic damage directly or indirectly via a variety of pathways. This injury can affect either somatic or germline cells, increasing the likelihood of cancer or inheritable defects ([Ambrożewicz et al., 2019](#)).

3.3.2.6 *Hemocompatibility*

Blood-compatible testing is carried out to verify the safety of materials used in blood-contacting devices (e.g., intravenous catheters, hemodialysis sets, blood transfusion sets, and vascular prostheses). Both products are in some ways incompatible with blood because they can disrupt blood cells (hemolysis), stimulate coagulation processes (thrombogenicity), and/or activate the complement system ([Poinern et al., 2014](#)).

3.3.2.7 *Preclinical safety testing*

Preclinical safety tests are employed in clinical development as well as before the start of human experiments to determine pharmacological and toxicological effects. Trials conducted in vivo as well as in vitro can help with this characterization ([Jiang et al., 2010](#)).

3.3.3 Different applications of biocompatible biopolymers

The use of biocompatible biopolymers in biomedical applications has drawn a lot of interest. Because biomaterials used in biomedical applications come into direct contact with or are associated with the human body, their safety is a matter of great concern. They must therefore be biocompatible, nontoxic, and fit the requirements for the intended use. As we have learned from the talks above, “biocompatibility” describes a biopolymer’s suitability for use in human body applications. For biomedical applications, biocompatible polymers possessing the three attributes listed below are appropriate ([Grenade et al., 2017](#)):

- **Hydrophilicity:** blood coagulation caused by unfavorable platelet adhesion is reduced by biomaterials possessing hydrophilic properties, making them useful in biomedical applications ([Grenade et al., 2017](#)).
- **Adhesion:** cell attachment should be a feature of biomaterials. Biomaterials expedite the potential for body tissue regeneration by taking on the role of

damaged tissue. Thus for these requirements to be met, interactions between cells and biomaterials are crucial (Eftekhari Ashtiani et al., 2021).

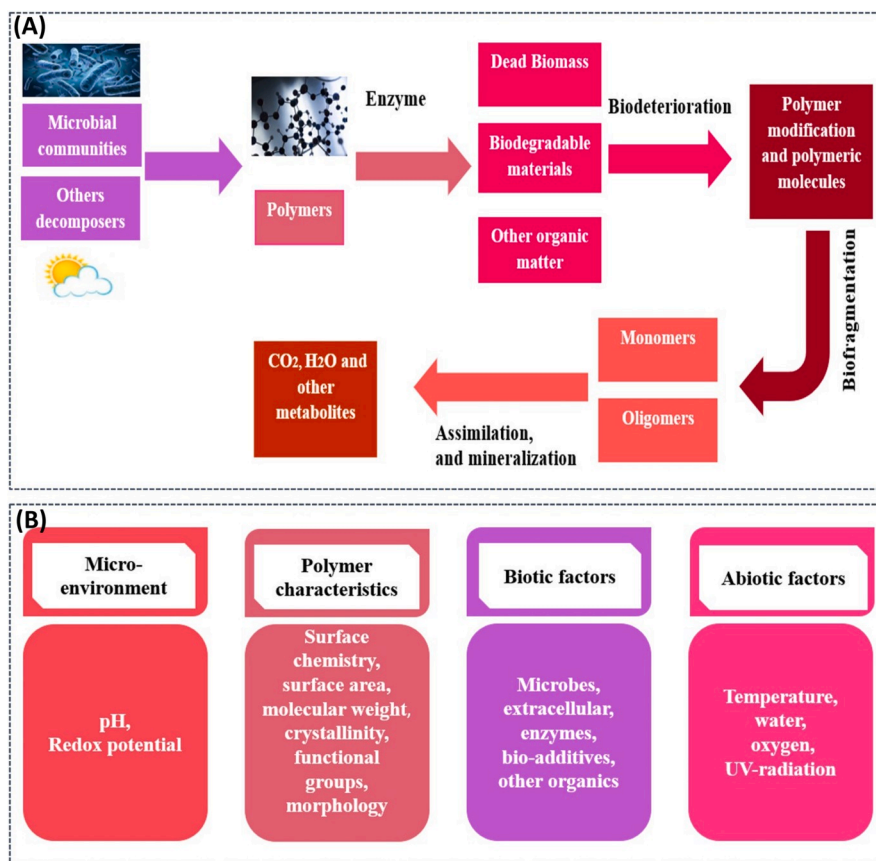
- Self-healing: the human body is designed to repair itself when it is injured, replacing damaged tissues. We refer to this process as self-healing. When a trauma or injury is too serious for the body to heal on its own, an external “material,” such as an implant, must be introduced. Thus the parts must have the ability to heal themselves (Pithon, 2017).

3.3.4 Biodegradability of biopolymers

Biodegradation is the process by which microorganisms, such as fungi, archaea, and bacteria break down organic material into carbon dioxide and H₂O under aerobic conditions. In the absence of complete mineralization, biotransformation occurs, resulting in the formation of organic and inorganic metabolites or transformation products (Jacob & Gopi, 2021; Kjeldsen et al., 2019; Li et al., 2020). In other words, the biodegradation of biopolymers is a microbial action in which biopolymers are partially or completely transmuted into end products (Zhou et al., 2022). The most important biodegradation processes for biopolymers, such as ester bond hydrolysis to liberate monomers, are carried out by extracellular enzymes excreted by microorganisms (Tokiwa et al., 2009). In general, polymer biodegradation occurs in four stages (Fig. 3.5A): biodeterioration, bio-fragmentation, assimilation, and mineralization (Harrison et al., 2018).

In the beginning phase, microbial biofilms form on the surface of polymers, and degraders and abiotic environmental factors break them down into particulate fragments, causing a change in the physicochemical properties of the polymers (Awasthi, Kumar, Kumar, et al., 2022). Extracellular enzymes are secreted from microbial biofilms in the second phase, specifically depolymerases, that transform smaller pieces of polymers into oligomeric, dimeric, and monomeric forms, resulting in a reduction in molecular size (Choe et al., 2021; Kjeldsen et al., 2019). In the following stage, oligomeric, dimeric, and monomeric forms are actively assimilated and utilized by microbes to produce cellular biomass and energy, as well as various groups of metabolites. Finally, the metabolites are mineralized/converted into CO₂, methane (CH₄), H₂O, and inorganic salts (Maddalwar et al., 2021; Zhou et al., 2022). Several factors influence the degradation of biopolymers (Fig. 3.5B), including polymer physicochemical properties and biotic and abiotic environmental factors (Nikolaivits et al., 2021).

Abiotic factors, such as humidity, O₂, temperature, sunlight, weathering, and pH, as well as biotic factors, such as indigenous microbial communities, work together to degrade biodegradable plastic polymers in the natural environment (Fuke et al., 2021; Thakur et al., 2018). The key degradation stage of polymers is abiotic hydrolysis, in which temperature and humidity facilitate the breaking of ester bonds (Karamanlioglu et al., 2017). When the environmental temperature rises above the glass temperature of biopolymers, the rate of hydrolysis and

**FIGURE 3.5**

(A) Schematic representation of biodegradation of the polymer and (B) factors affecting the degradation of the biodegradable polymers.

biodeterioration increases (Choe et al., 2021). Similarly, the availability of O₂ regulates the diversity of bio-degraders and regulates biological reactions. Aerobic microbes use plastic polymers as sources of energy and carbon in the presence of O₂ (Sudesh & Iwata, 2008). Anaerobic microbes degrade plastics and polymers in anaerobic environments and produce biogas, primarily in the form of CH₄ (Bátori et al., 2018). Biodegradation in a variety of environments, such as soil, H₂O, and compost, has been studied using isolated strains of bacteria and/or fungi (Paiva et al., 2018). When biopolymers are released into the environment after being manufactured, they degrade rapidly due to microbial enzymatic activities. CO₂, CH₄, H₂O, humic matter, and a variety of other naturally occurring chemicals are

discharged into the surrounding environment during the breakdown of organic matter (OM) (Awasthi, Kumar, Sarsaiya, et al., 2022; Rai et al., 2021). As a result, the use of bio-based and biodegradable polymers that can degrade and be managed in natural environmental settings; a global movement is required to combat both plastic pollution and climate change, both of which are critical for achieving the goals of sustainable development and lowering the carbon footprint (Meereboer et al., 2020).

3.3.5 Biocompatibility and biodegradability for specific biopolymers

Chitosan, the second largest natural biopolymer after cellulose, is a cationic polysaccharide extracted from chitin via deacetylation and composed principally of units that repeat of β -(1,4)-2-amino-2-deoxy-D-glucose (or D-glucosamine) (Dutta et al., 2004; Kumar et al., 2010). It possesses unique chemical and physical characteristics, including nontoxicity, biodegradability, biocompatibility, renewability, and antibacterial activity (Li & Tang, 2016; Wieckiewicz et al., 2016). In many plant species, chitosan can have a significant impact on growth regulation and defense induction. However, it is still unclear how precisely plants react metabolically to chitosan (Zhang et al., 2017). Additionally, it has been sold all over the world as a fat binder in formulations meant to lower cholesterol and slim down, as well as an ingredient in nonmedical products (Shahidi et al., 1999). Due to its cationic nature, chitosan has been found to entrap lipids in the intestine (Kanauchi et al., 1995; Wuolijoki et al., 1999). It has been shown through biomedical analyses to be extremely biocompatible (George & Abraham, 2006; Muzzarelli et al., 1988). Simultaneously, chitosan is metabolized by certain human enzymes, particularly lysozyme, and is considered biodegradable. Chitosan possesses distinct biological properties, such as biocompatibility, biodegradability, and mucoadhesion. It is also anticholesterolemic, antimicrobial, and has permeation-enhancing properties (Berger et al., 2004; Rinaudo, 2006). Due to these features, it is increasingly being used in a variety of applications, including antibacterial/antibiofouling coatings, controlled release coatings and microcapsules, nanofiltration, drug delivery hydrogels, gene delivery, and tissue engineering scaffolds. Chitosan derived from nonanimal origins is preferred for biomedical applications aiming for in vivo testing (Balaji et al., 2017; Liu et al., 2019; Nasrollahzadeh et al., 2020).

3.4 Life cycle assessment and toxicity of biopolymers

LCA is a method and approach for assessing the environmental impact of a product, from the production of raw materials to the recovery and generation of waste (Mármol et al., 2023; Romeiko et al., 2024). LCA involves the collection and

collation of information and data relating to the inputs and outputs, including emissions, residues, and resources, of a process (life cycle inventory), and the conversion and translation of this data into ecological consequences (via impact measurement methodologies), such as participation in and contribution to global warming, smog creation and formation, eutrophication, acidification, or harmfulness and toxicity to humans and ecosystems (He et al., 2024). After normalization, it is possible to classify and present the results by impact category, as well as grouping and dividing them according to damage categories (e.g., “human health,” “ecosystem quality,” and “resources,” which are involved in the Ecoindicator-99 methodology). This standardization consists of distributing and dividing the scores obtained in the impact categories according to a person’s average annual participation and contribution, to be able to combine different categories whose scores now have no units. As some environmental effects appear to be more serious and far-reaching than others, a weighting of the standardized and normalized impact score can also be applied and assigned. However, the choice and relative decision of the importance of environmental impacts is subjective and therefore not based on any scientific evidence (Samani, 2023; Standardization, 2006a), and the methods used and proposed do not meet with consensus. LCA standards have been established to a certain extent within the framework of the International Organization for Standardization, but these guidelines leave a great deal of room for maneuver and interpretation for the LCA specialist and practitioner. As a result, comparing results from different studies is sometimes difficult and, in any case, requires a great deal of care for a variety of reasons, including methodological differences (such as functional unit, allocation method, and impact assessment method) and the inventory data involved. The life cycle of petrochemical polymers begins when the required raw materials, particularly fossil raw materials, become available (Al et al., 2024). Feedstocks are combined with (nonrenewable) energy requirements to determine and calculate the depletion rate of fossil fuel-based fuels, usually presented as National Renewable Energy Unit. To do this, we need to consider the starting material as energy, rather than as a material input, which means transforming the quantity used into the heat of combustion (Standardization, 2006b). Plastics-Europe provides an LCI database for petrochemical plastics based on European averages and values, which has been used in most studies and comparative analyses with biopolymers. LCAs for biopolymers derived from agricultural resources include the cultivation of the plant in which the biopolymer is used. This includes the fuel required for cultivation operations (plowing, application of agrochemicals, harvesting, etc.). The manufacture and transport of the products and materials required, such as fertilizers, herbicides, and pesticides, should also be considered (Mármol et al., 2023). H₂O consumption and land management are also key factors, as are nitrogen emissions from the manufacture and use of fertilizers. Other processes to be integrated for conventional biopolymers include milling and production.

Indeed, some studies use life-cycle analysis to distinguish and compare biopolymers from the petrochemical plastics they are likely to replace. In some

cases, these are complete life-cycle assessments, from start to finish, while in others, only the impacts up to the point where the polymer granules or certain products leave the plant are considered (these are referred to as eco profiles). Although studies focusing solely on the manufacture of granules or products have the advantage of being more easily comparable with other analyses, the omission of downstream processing can be significant and lead to a change in the preferred material depending on its use and its circuit and mode of disposal. For product-type eco-profiles and integral LCAs, the manufacture of the part and item in question must be added (Mármol et al., 2023). This stage has generally been omitted from comparative studies, as the environmental impact was assumed to be equivalent for all polymers, provided the same processes were used. In addition, phases involving use and, less frequently, transport were neglected in the “start-to-finish” studies. However, if the mass requirements for a product differ according to the material used, this has been considered. Cradle-to-grave studies also include the effects of end-of-life processes. The options considered include landfill, incineration, industrial and domestic composting, anaerobic digestion, and recycling.

3.5 Social economic aspects

Our society has grown because of the extraction and use of natural resources. However, many natural resources are running low, and if human activity keeps up its current pace, more than one planet will be needed to support human consumption and production. Our society must use natural resources more wisely to strike a balance between resource consumption and economic growth (Hottle et al., 2017). Motivating society to adopt an environmentally conscious mindset is essential to fully implementing the concept of sustainable development in a market economy. This kind of information is essential for selecting environmentally friendly products that the market will accept quickly and for developing chances to control consumer motivation toward an environmentally friendly product (Competition & Works, 2021). Thus far, very few studies have examined the economic aspects of recycling waste biopolymers or bioplastics. This is due to several valid reasons. Less than 1% of the world’s plastics market is made up of biopolymers, which means that they lack economies of scale and do not yet have sufficient waste streams. Second, there is no need to recycle biopolymers because it is widely believed that they are compostable or biodegradable (Niaounakis, 2013b). However, pollution can be decreased even when recycled materials are used. People’s behavior must be encouraged to adapt to these changes. Utilizing a variety of initiatives is required to encourage individuals to use fewer nonenvironmental resources. This method is more effective than individual approaches (Borel-Saladin & Turok, 2013). Furthermore, recycling biopolymers is unnecessary because it is widely believed that they are compostable or biodegradable. However, most biopolymers only break down under circumstances, like in industrial composting, rather than spontaneously

in the environment. Furthermore, not all biopolymers can be composted because they can be classified as either compostable polymers derived from fossil sources or renewable polymers that are not necessarily biodegradable (Niaounakis, 2013a). Achieving the highest level of waste recovery and recycling and then reusing it in production is the socio-economy's technological objective. Biopolymers, which decompose naturally and produce byproducts that do not pose a health risk to humans or animals, are a potential replacement for conventional and petroleum-based materials (Wojnarowska et al., 2021). To ensure the successful implementation of this environmentally friendly idea, an economic analysis before and after the switch from traditional petroleum-based polymers to biodegradable synthetic biopolymers is required. This analysis will assist in determining the potential impact on consumer health and safety, as well as the environmental and economic implications (Stoica et al., 2020). Biopolymers have a wide range of social applications. Nowadays, biopolymers are used in daily life. For instance, biopolymers are utilized in solar cells, electrical instruments, cooking utensils, medical, agricultural, hydrogen storage, space research, packaging, and corrosion protection applications, and also they are used more often in the environment; they are an adsorbent for heavy metals and toxic ions and are used in the purification of H₂O (Nagaraj & Rajan, 2020). When analyzing the social factor that determines the exposure to and effects of toxic substances during the production of biopolymers, the human health indicator is taken into account (Hertwich et al., 2001). To assess the economic feasibility of the biopolymer, three indicators are chosen: production cost, market price, and global production capacity. Product expenses, such as capital, maintenance, and operational costs are included in the production cost (Alex Marvin et al., 2012). The market price reflects the economic value of the biopolymer, which is determined by supply and demand forces (Mtibe et al., 2021). Global production capacity depicts biopolymer production capacities around the world, which are used to calculate growth rates and emerging trends in biopolymers (Galiano et al., 2018). One of the primary goals of the past few decades has been the global transition to sustainable development, which includes developing national and regional bioeconomy strategies (Heimann, 2019). The bioeconomy encompasses all sectors and systems that rely on biological resources, such as animals, plants, microorganisms, derived biomass, and organic waste, as well as their functions and principles (Sanz-Hernández et al., 2019). The bioeconomy provides a chance to address plastic pollution, one of the world's issues. The take-make-dispose cycle is what the linear economy theory advocates. It leaves a long-term trace in the atmosphere, soil, ocean, animal biomass, economy, and society (Economics et al., 2020). Therefore the eco-friendliness of the environment, health benefits, affordability, lower operating costs, and other factors should be among the socio-economic benefits of biopolymers that most appeal to end users. The potential for profit interests consumer producers as well. State institutions are interested in environmental goods that contribute to the nation's competitiveness, health, and economic and environmental security as well as the growth of the national economy and state budget. A few distinct features of environmental goods, like resource

conservation and zero emissions during operation and disposal, are of interest to consumers like foreign companies and unions. More to the interests of intermediaries are factors like price, potential profit margin, sales growth, etc. ([Competition & Works, 2021](#)).

3.6 Environmental and health impact of biopolymers

3.6.1 Environmental impact of biopolymers

Due to their biodegradable character, their inoffensive, and environmentally friendly production processes and techniques, as well as their multiple applications and uses, biopolymers are interesting and viable alternatives to nonrenewable materials and products. Synthetic polymers have damaged our ecosystem. Any attempt to derive materials from renewable sources that are likely to degrade very rapidly in the environment is therefore essential to remedy and repair the damage caused by their excessive use and prevent further deterioration. The demand for biopolymers, such as lipids, polysaccharides, and proteins as renewable raw materials and sources to replace nonbiodegradable, petroleum-derived plastics has increased in recent years ([Elgarahy et al., 2023](#); [Rendón-Villalobos et al., 2016](#)).

Producing polymers from renewable and natural sources has a dual effect on the environment. Firstly, it is possible to renew the raw materials used, whether through natural cycles or intentional and voluntary human action and intervention. Secondly, the biodegradability of derived products represents another ecological advantage of using sustainable raw materials for biopolymer production, thus preventing the potential contamination associated with the destruction of a corresponding volume of traditional plastic products. These biopolymeric materials are generally disposed of in landfills or recycled when they reach the end of their useful life. Synthetic materials are recycled and enjoy a high profile, but efforts to develop this type of recycling have met with little success. It should be recognized that recycling is a method of waste treatment, not a goal of materials development. Complacency about recycling processes ignores the fact that a large infrastructure is essential to ensure that recycling operations run smoothly ([Aher et al., 2023](#); [Burzic et al., 2019](#); [Shamsuddin et al., 2018](#); [Udayakumar et al., 2021](#)). As Mulder discovered, plastics are practically recycled, since the return on investment in their economic situation is favorable ([Burzic et al., 2019](#); [Mulder, 1998](#)). Recycling appeared at first to be an efficient and cost-effective method of reducing pollution and environmental damage, as a residue management process. However, as time has gone by, it has become clear that the use of plastic materials from sustainable, biodegradable sources represents a more sensible alternative to conventional plastics processing and recycling since the materials used constitute OM that avoids the emission of harmful products. This is why it is essential to promote the development of composable or easily biodegradable plastics. When biopolymers are disposed of in landfill sites, the microorganisms required are expected

to be present. However, unless the soil is inoculated with them, this is not always the case. Inoculations with bacteria, fungi, and actinomycetes are useful for the decomposition of biopolymers within the soil. However, in the presence of the micro-organisms concerned, the destruction and disposal (and subsequent decomposition) of biodegradable (or partially biodegradable) plastic materials will increase the capacity and space available in existing landfills, as the volume of residues is reduced through biodegradation.

Bio-based compostable synthetics degrade during the recycling process to generate carbon dioxide, H_2O , mineral compounds, and biomass at a rate consistent and compatible with other existing compostable materials, without leaving any visible or toxic waste. To be integrated into industrial composters, these biopolymers require complete and total biodegradation and disintegration, as well as the absence of consequences of biopolymer degradation in terms of substrate quality. An additional benefit is the reduction of odor emissions from compost heaps when biodegradable plastics are incorporated into the compost composition and mix. The decomposition of compost generates ammonia, a noxious gas. The degradation process of biodegradable plastics generates acid intermediate products that neutralize the ammonia level, thus resolving any odor-related problems (Kumar et al., 2023; Moshood et al., 2022). Indeed, despite the enormous advantages of biopolymers over plastics, there are a few potential adverse environmental effects: (1) waste and environmental contamination, (2) source and production of raw materials, (3) energy requirements, (4) involvement of catalysts, (5) presence of solvents, and (6) presence of stubborn residues. Some biopolymers, such as PLA, which biodegrade gradually in the natural environment, run the risk of polluting the soil, entering rivers and streams, and eventually ending up in the sea. The biodegradation process of biopolymers can also have a certain impact and effect on the characteristics and concentrations of soil microflora in landfill sites. In general, biopolymers are obtained from ethanol, using corn and sugar as raw materials. The production of ethanol-based biopolymers also appears to have the same problems as those associated with biofuels. In general, bio-based commodities are grown using agro-industrial production techniques and, as a result, large quantities of energy, H_2O , land, genetically modified organisms, pesticides, and toxic fertilizers are employed. For optimum economic performance, monomers usually must undergo polymerization with certain metal catalysts. PLA polymerization uses tin catalysts. In the case of biodegradable biopolymers, these residual catalysts are released and may be disposed of in the environment. Most biodegradable biopolymers are produced from plant raw materials. These are then isolated from the solvent used for extraction through a range of purification processes (Ashter, 2016; Burzic et al., 2019; Kumar et al., 2023; Moshood et al., 2022; Shamsuddin et al., 2018).

3.6.2 Health impact of biopolymers

Furthermore, in the context of an LCA, Alvarez-Chavez et al. have carried out a qualitative assessment of the adverse effects of biobased polymers on human health

and the environment (Álvarez-Chávez et al., 2012; Ashter, 2016; Udayakumar et al., 2021). Polyhydroxyalkanoates (PHA), PLA, cellulose, lignin, hemicellulose, and starch are recognized as biopolymers that do not present any risk of causing diseases or health disorders. Each environment entails different risks. Military personnel have always been exposed to hazards, either in the workplace or in theaters of operation, with adverse consequences for their health. From the beginning, the main aim of military medical services was to reduce or eliminate injuries caused by malnutrition, improper fluids and waste, infections, climatic conditions, equipment worn by the soldiers, etc. However, recent studies have revealed a reduction in the risks associated with occupational exposure and environmental exposure to harmful chemicals, potentially toxic substances, and common pharmaceutical and chemical products intended for military use. However, recent studies reveal a reduction in the risks associated with occupational and environmental exposure to harmful chemicals, potentially toxic substances and common pharmaceutical and chemical products intended for military use. In view

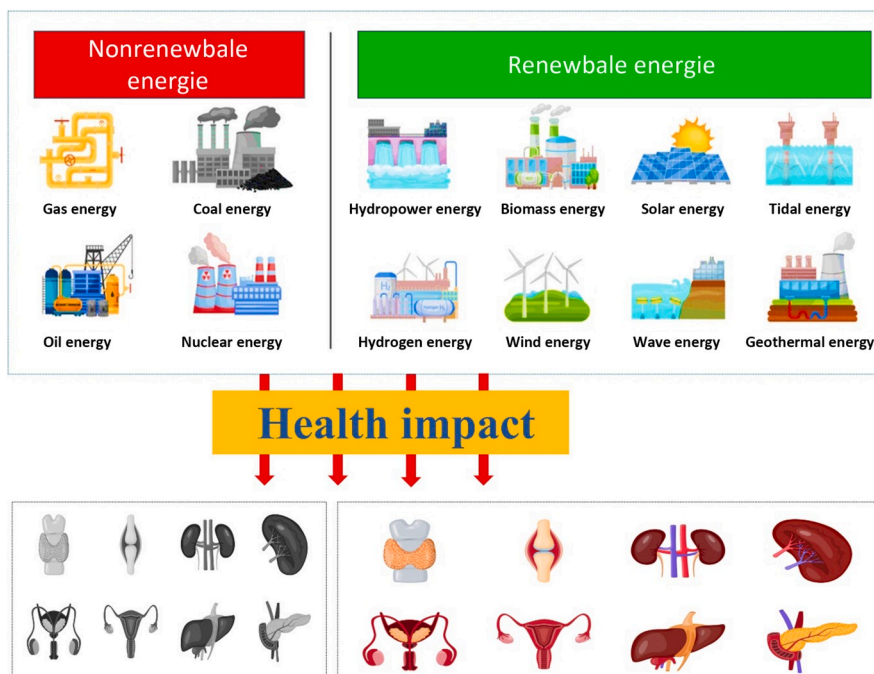


FIGURE 3.6

Representative diagram of the impact of different energy sources on human health and organisms.

of current regulations and the resulting concerns, the potential for military personnel to consider changing the composition of munitions in order to reduce or eliminate the harmfulness ratio for both human health and the environment must be taken into account (Ashter, 2016; Das et al., 2023; Zecheru, 2010). In general, biopolymers have no direct harmful effects on human health since they are usually derived from natural resources and are biodegradable (Fig. 3.6). However, the only issue that may indirectly affect human health, in particular skin, hair, etc., is the method and process used to isolate, extract and produce the various biopolymers (e.g., cellulose, lignin, hemicellulose, tin, starch, etc.) (Fig. 3.6).

3.7 Conclusion

Over the last few decades, industrial interest in biopolymers and green products derived from biopolymers has grown steadily. These materials are renewable, harmless, ecological, environmentally friendly, biodegradable, lightweight, very easy to fabricate, and possess a significant structure that gives them improved mechanical strength and good toughness, and they can be adapted and modified to meet and satisfy the needs of various performance requirements. Several criteria determine the properties of biopolymer-based hybrid materials, including chemical composition, service life, environmental impact, particle size, and surface area. This in-depth study revealed that biopolymers are suitable for use in tissue technology, drug packaging, agricultural and pharmaceutical products, biomedical products, bioengineering, and food production, as well as a wide range of intervention possibilities. The most important issues facing humanity in the future concern energy and resources, food, health, transport and infrastructure, and communication. Biopolymers are essential to solving these problems. Biopolymers will be the materials of the next century, and their production will ensure global access to the best solutions. Synthetic biopolymer technology has long been used in medicine. In particular, they are used in the manufacture of medical equipment and artificial organs. The medicine of the future may also involve synthetic polymers. The polymer sector saves energy and promotes and stimulates renewable resources. In the future, fossil fuels will not be sufficient to meet the world's energy needs. Fuel cells are planned to facilitate the realization of this project. They can replace electric batteries and internal combustion engine generators. The value of biopolymers is set to increase through the development of many new adaptations of solid materials and the continued reduction in the cost of producing these plastic-based biomaterials. Bioplastics have the potential to replace conventional plastics in a wide range of applications, including food packaging, plastic plates, cups, blankets, and storage packaging, thereby helping to limit the impact of human activity on the environment. Further research into biopolymer-based materials has the potential to replace harmful, nonbiodegradable traditional materials in the very near future. The improved durability and performance of these bioproducts pave the way for the

creation of new markets to replace petroleum-based products with nontoxic, highly durable, and environmentally friendly materials. Ongoing research into the benefits, performance, and life-cycle assessment of biopolymers is essential to bring the notion of sustainability into play.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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